Cross-Scale-Guided Fusion Transformer for Disaster Assessment Using Satellite Imagery

Weiwei Xiao, *Graduate Student Member, IEEE*, Jingyong Su, Yongyong Chen, *Member, IEEE*, and Guofeng Cao

***Abstract*— When a disaster strikes, accurate disaster information and effective response are critical for saving lives and properties. High-resolution satellite (HRS) imagery provides valuable geographical information that can assist experts in analyzing damage levels in different areas and enacting appropriate relief plans. However, analyzing large HRS images is both time-consuming and inefficient, requiring efficient automated methods to replace expert analysis. Fortunately, deep learning methods have achieved impressive performance on HRS image processing tasks, considerably increasing automation levels. Despite this progress, most HRS-based damage assessment methods only consider a single time series of postdisaster images or simply integrate pre- and postdisaster images, lacking the integration of effective information between pre- and postdisaster images. To alleviate this problem, we propose a two-stage multiscale fusion network that fully exploits the information contained in pre- and postdisaster images. Specifically, we employ a hierarchical transformer to accurately locate buildings by predisaster images in the first stage, and then propose the guided fusion (GF) and cross-scale GF modules in the second stage to efficiently utilize both pre- and postdisaster images. Our method outperforms state-of-the-art (SOTA) methods in building segmentation and building damage assessment on the xBD dataset, and exhibits improved generalization across diverse geographic regions and disaster types.**

***Index Terms*— Cross-scale-guided fusion (CSGF), damage assessment, satellite imagery, siamese transformer.**

I. Introduction

With the increasing occurrence of the extreme weather events caused by climate change, natural disasters such as earthquakes, tsunamis, and typhoons have become more frequent in recent years, resulting in tens of thousands of lives lost and huge economic losses [1]. For instance, Hurricane Harvey in the USA in 2018 claimed 107 lives and resulted in economic losses of up to 125 billion U.S. dollars [2]. Responding quickly to an emergency after a disaster is critical to ensuring personnel safety and reducing loss of property. The continuous advancement of remote sensing technology has led to the emergence of numerous disaster assessment methods on HRS images [3], [4], [5], [6]. However, traditional methods primarily compare pre- and postdisaster images by handcrafted features according to change detection [7], [8], which are not well-suited for analyzing multiple natural disasters.

Deep learning-based methods have achieved remarkable success in HRS image processing [9], [10], leading to the emergency of numerous building damage assessment methods. Some building damage assessment methods rely solely on postdisaster images, which cannot accurately identify the location of building areas [11], [12], [13]. In contrast, other methods are based on change detection using pre- and postdisaster images to obtain the location of buildings and assess the damage level of buildings. Similar to change detection, building damage assessment primarily relies on identifying differences between pre- and postdisaster images. Therefore, these methods mainly focus on fusion strategies of images before and after changes. As illustrated in Fig. 1, based on the fusion position of images, three categories have been identified: early fusion [14], late fusion [15], and middle fusion [16], [17], [18]. Early fusion and late fusion lack a multilevel

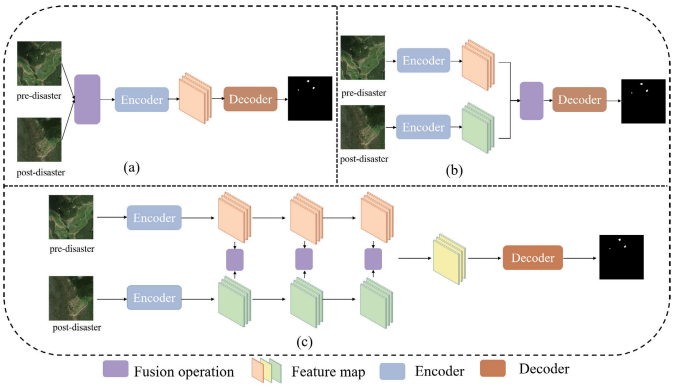


Fig. 1. Different fusion strategies of pre- and postdisaster images in deep learning methods. (a) Early fusion strategy fuses the input before the network. (b) Late fusion strategy fuses the output of the encoder. (c) Middle fusion strategy fuses the features of the middle layer.

Manuscript received 11 June 2023; accepted 13 July 2023. Date of publication 31 July 2023; date of current version 7 August 2023. This work was supported in part by the GuangDong Basic and Applied Basic Research Foundation under Grant 2022A1515010800, in part by the GuangDong Natural Science Foundation under Grant 2022A1515010819, in part by the Shenzhen Science and Technology Innovation Program under Grant JCYJ20220818102414031, in part by the National Natural Science Foundation of China under Grant 62106063, and in part by the U.S. Department of Commerce National Institute of Standards and Technology (NIST) under Grant #70NANB19H061. *(Corresponding authors: Jingyong Su; Guofeng Cao.)*

Weiwei Xiao and Jingyong Su are with the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China, and also with the Peng Cheng Laboratory, Shenzhen 518066, China (e-mail: xiaowwhit@gmail.com; sujingyong@hit.edu.cn).

Yongyong Chen is with the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China (e-mail: YongyongChen.cn@gmail.com).

Guofeng Cao is with the Department of Geography, University of Colorado Boulder, Boulder, CO 80309 USA (e-mail: guofeng.cao@colorado.edu).

Digital Object Identifier 10.1109/TGRS.2023.3298037

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

fusion strategies, while middle fusion provides more comprehensive information interactions. The fusion strategy for building damage assessment typically involves concatenation [15], attention [18], and others, which can partially improve the location detection of changed regions. However, there are two fundamental differences between damage assessment and change detection. First, building damage assessment must detect all building areas, including both the changed or not regions. Second, building damage assessment must estimate multiple levels for each damaged or not region.

To address these challenges, we adopt a two-stage method for building damage assessment. First, as shown in Fig. 2, the severe damage to building areas in postdisaster images hinder the ability to obtain clear building outlines. Therefore, we use only predisaster images in the first stage to locate buildings. Then, we observe that middle fusion methods, such as [16], [17], [18], lack multiscale fusion. To overcome this limitation, we propose the guided fusion (GF) module and the cross-scale GF (CSGF) module in the second stage, leveraging a dual-branch hierarchical transformer network in the encoding process. Specifically, we introduce the GF module to transfer the deep features to the shallow features, and the CSGF module to enable weighted fusion of features at different scales. To benchmark our method, we compare it with existing state-of-the-art (SOTA) methods on the publicly available large-scale damage assessment xBD dataset [19], which contains HRS images of various natural hazards, such as earthquakes, floods and hurricanes etc. Our contributions are as follows.

1) We propose a novel two-stage method for building damage assessment, which first uses hierarchical transformer structure to locate buildings in predisaster images, effectively addressing the challenge of identifying building areas in severely damaged postdisaster images.
2) We propose the GF and CSGF modules in the second stage enhance the multiscale fusion process by transferring deep features to shallow features and enabling weighted fusion of different scale features, respectively.
3) Our proposed method achieves SOTA performance not only in building segmentation but also in building damage assessment on the xBD dataset.

The remainder of this article is organized as follows. Section II provides an overview of related work on building damage assessment. In Section III, we describe our model architecture and provide a detailed explanation of the GF and CSGF modules. Section IV presents an analysis of experimental results on xBD dataset, and Section V concludes the article.

II. Related Work

In this section, we first introduce the task of building damage assessment and review related work. Then, we discuss the transformer structure and feature fusion methods, from the perspective of building segmentation and damage assessment, respectively.

A. Building Damage Assessment

Building damage assessment primarily comprises two subtasks: building segmentation and damage assessment. Building segmentation aims to accurately locate building regions, while damage assessment classifies the severity of damage based on their locations. These two subtasks are closely related, as accurate building location improves the performance of damage assessment.

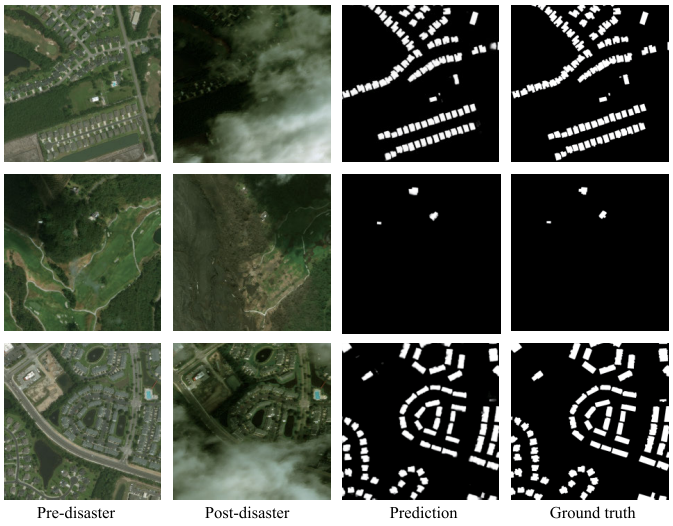


Fig. 2. Some examples of pre- and postdisaster HRS images.

A portion of the research in building damage assessment has focused on evaluating damage of known building locations. For instance, Cao and Choe [20] proposed a disaster assessment method that combines HRS images with some prior geographic features to transform the damage assessment task into a multiclass classification task for known building locations. However, this known building location methods impose serious limitations on building damage assessment scenarios. Another part of the research mainly deals with cases when the building location is unknown. In this scenario, building damage assessment methods aim to directly extract the building location from HRS images and assess the building damage. These unknown building location methods can be divided into two main categories: the one-stage method and the two-stage method.

The one-stage method has its origins in building change detection task [21], where two HRS images captured at different times are compared to identify changes in buildings and surrounding areas. In contrast to change detection task, damage assessment involves identifying the extent of changes in building areas and estimating the severity of building damage. Based on the change detection framework, several studies have proposed one-stage damage assessment methods. Bai et al. [22] proposed a pool pyramid model that fuses different receptive field features to assess the extent of building damage. Weber and Kané [15] proposed a method that incorporates deep features of pre- and postdisaster images, significantly outperformed the baseline model in the xView2 challenge.[1] Hao et al. [18] proposed a siamese U-net model with attention mechanism, which fuses the pre- and postdisaster image to improve damage assessment performance. Xie et al. [17] proposed a method that utilizes a feature

[1]https://xview2.org

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

fusion module and contrastive learning to improve damage assessment performance.

The two-stage methods, as the name suggests, divide building segmentation and damage assessment into two separate stages. Gupta et al. [19] proposed a two-stage baseline model for building damage assessment and released a large high-resolution satellite (HRS) image dataset, xBD. The baseline model uses a segmentation model in the first stage to extract the location of building and a classification model in the second stage to assess the damage extent in the building area. Shen et al. [23] proposed a two-stage method for building damage assessment, which utilizes a U-Net model for building segmentation in the first stage and employs a dual-branch network with a cross-attention mechanism in the second stage to fuse pre- and postdisaster images. Deng and Wang [16] introduced an attention mechanism to weight the fusion of pre- and postdisaster images in the second stage.

In general, the one-stage methods can achieve end-to-end training but may yield lower building segmentation results than two-stage methods. Both one-stage and two-stage methods aim to obtain accurate building locations and design fusion mechanisms between pre- and postdisaster images.

B. Transformer

Transformer [24] was proposed by Google team in 2017 and achieved remarkable performance in machine translation. With transformer structures excelling in natural language processing (NLP) tasks, researchers have also turned their attention to applying them in computer vision. In 2020, the Google team introduced a distinct transformer structure for computer vision tasks, called ViT, and achieved first in the ImageNet [25] classification competition. The self-attention mechanism between different patches in transformer-based models has led to excellent performance in various visual tasks, such as object detection [26], semantic segmentation [27], and super-resolution [28].

Since damage assessment can be tackled with semantic segmentation, we are particularly interested in exploring transformer-based segmentation methods. SETR [29] proposed the first transformer-based semantic segmentation method, which splits images into patches and uses self-attention mechanism to obtain context information between patches, unlike traditional methods like FCN and U-Net that decrease resolution through encoder to obtain a larger perception field. Segformer [27] proposed a lightweight transformer encoding structure with hierarchical features to improve reasoning time, which is distinct from SETR's method of maintaining a constant feature scale. Swin-Transformer [30] proposed a hierarchical transformer with sliding window operations to handle the challenge of increased computational cost caused by high-resolution images and the detection of objects at various scales. PVT [31] proposed a pyramid transformer structure, which has been successfully adapted to various dense prediction tasks by stacking multiple independent transformer encoder structures to obtain encoded features at different resolutions.

With the continuous improvement of the transformer structure, the performance of transformer outperforms the CNN-based methods in most computer vision tasks. There is a trend for transformer to gradually replace CNN as the dominant encoding structure.

C. Feature Fusion

Image fusion can be divided into three levels: pixel-level, feature-level, and decision-level, depending on the level of image representation. This section focuses on the feature-level fusion, which involves the fusion of features extracted by the feature extractor. Feature fusion not only increases the diversity of information but also generates complementary and complex feature.

Traditional methods for feature extraction rely on operators such as histogram of oriented gradient (HOG) [32], scale invariant feature transform (SIFT) [33], and oriented fast and rotated brief (ORB) [34], followed by feature fusion through techniques such as stitching or Laplacian pyramid. In contrast, deep learning methods are capable of obtaining hierarchical features of different scales through deep neural networks, providing a more advanced approach to feature extraction and fusion. Feature fusion can be divided into two categories based on the scale of features: same-scale fusion and cross-scale fusion. Various methods exist for same-scale fusion, including feature addition, matrix multiplication, and feature concatenation. In contrast, cross-scale fusion methods are more diverse, ranging from network structure fusion to fusion modules. Representative examples of network structure fusion are ResNet [35] and DenseNet [36], which transfer features from shallow to deep layers and mitigate the vanishing gradient problem. Specific fusion modules for cross-scale feature fusion include feature pyramid network (FPN) [37], pyramid scene parsing (PSP) [38], and spatial pyramid pooling (SPP) [39].

Essentially, the attention mechanism can be viewed as a weighted feature fusion method that focuses on high-weight items and disregards irrelevant ones, thus improving the network performance by adapting the weights. SPANet [40] proposed a spatial pyramid attention mechanism that captures multiscale information through multiple pooling kernels of different sizes, similar to SPP [39].

III. Method

We begin with an overview of our approach, which involves building location segmentation and damage assessment, as illustrated in Fig. 3. This section provides a comprehensive description of our proposed two-stage method, as well as the two fusion modules proposed in the second stage.

A. Two-Stage Structure

The primary concept behind our two-stage method is to decouple building damage assessment into two distinct stages: building segmentation and damage assessment.

1) Stage 1 (Building Segmentation): While the self-attention mechanism in transformer structures allows for strong global feature extraction capabilities, it lacks fine-grained semantic information. In contrast, the use of multiscale convolution kernels in CNN can efficiently extract fine-grained semantic information, but may not capture the broader

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

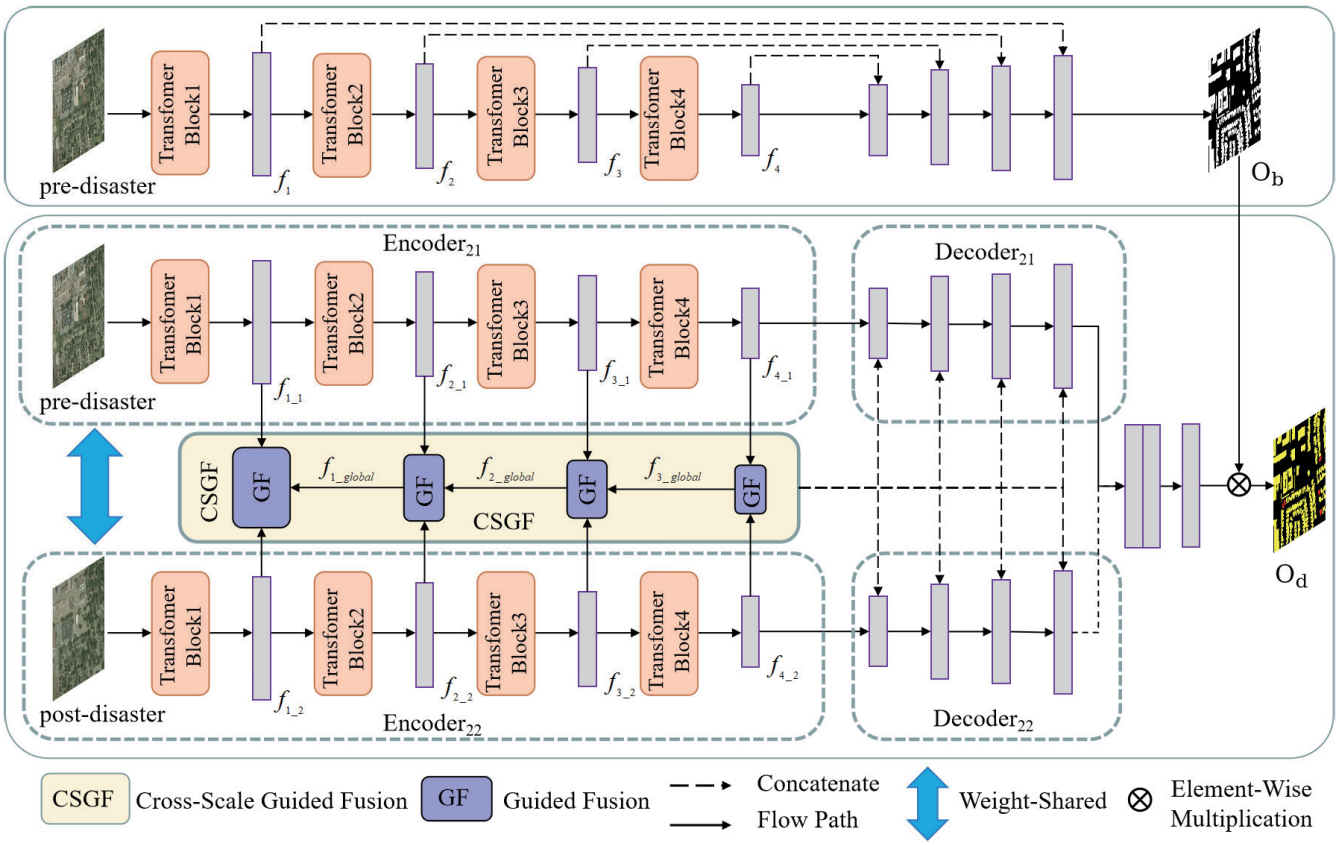


Fig. 3. Proposed two-stage building damage assessment framework. In the first stage, our method employs a hierarchical transformer to segment the predisaster image and extract clear building locations. In the second stage, a dual-branch transformer structure is utilized to process both pre- and postdisaster images. Within the dual-branch network, the GF module is employed to transfer deep features to shallow layer (refer to Fig. 4 for specific details), while the CSGF module is employed to perform weighted fusion of multiscale features (refer to Fig. 5 for details). These two modules enable our method to effectively leverage both pre- and postdisaster images, facilitating more accurate building damage assessment.

contextual information. Transformer combined with CNN has achieved impressive results in medical image segmentation. Trans-U-Net [41] combines transformer with U-Net to significantly improve the performance of medical image segmentation.

As shown in Fig. 3, our method employs a hierarchical transformer structure as the encoder to extract features in the first stage, with the encoder utilizing the Segformer [27] structure as the backbone. Instead of using the decoder structure of multilayer perception (MLP) in Segformer, we employ a multiscale CNN structure to decode the hierarchical features. Specifically, our method utilizes a connection relationship similar to that of U-Net between the encoder and decoder. Since predisaster images often provide clear depictions of building outlines than postdisaster images, our method trains the building segmentation model primarily using predisaster images in the first stage. After the completion of the segmentation model training, we utilize the encoder weights as the initial parameters for the second-stage model. Moreover, the building segmentation results from the first stage can improve the accuracy of the building damage assessment in the second stage. The outputs of the segmentation network in the first stage are as follows:

$$O_b = \arg\max_c \left(O_p(c,h,w)\right) \tag{1}$$

where $O_p \in \mathbb{R}^{2\times H\times W}$ represents the output of the network and $O_b \in \mathbb{R}^{H\times W}$ represents the final predicted result.

During the building segmentation stage, a significant class imbalance arises due to the large discrepancy in the number of pixels between building areas and others in HRS imagery. In order to alleviate this problem, we employ focal loss [42] to dynamically adjust the weights of positive and negative samples during the training process. The specific loss function is as follows:

$$L_{\text{focal}}\left(y_i, \hat{y}_i\right) = \begin{cases} -\left(1-\hat{y_{il}}\right)^{\gamma}\log\left(\hat{y_{il}}\right), & \text{if } y_{il}=1 \\ -\left(\hat{y_{il}}\right)^{\gamma}\log\left(1-\hat{y_{il}}\right), & \text{if } y_{il}=0 \end{cases} \tag{2}$$

where y_{il} represents the binary label of ground-truth y_i indicating whether the current pixel belongs to a building or not, and $\hat{y_{il}}$ represents the binary label of predicted values $\hat{y}_i$. γ is a hyperparameter to reduce the loss of samples that are easily distinguishable.

Additionally, to enhance the precision of building boundary prediction, we also employ dice loss [43] for building segmentation

$$L_{\text{dice}}\left(y_i, \hat{y}_i\right) = 1 - \frac{2\sum_{i=1}^{N} y_{il}\hat{y_{il}}}{\sum_{i=1}^{N} y_{il} + \sum_{i=1}^{N} \hat{y_{il}}}. \tag{3}$$

Due to the binary nature of the mask tag constructed from y_{il}, the numerator term can be computed as the product of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETERS OF OUR METHOD

Stage 1			Stage 2			
Layer	Output Size	Block	Layer	Layer	Output Size	Block
Input pre-disaster	$512 \times 512 \times 3$	-	Input pre-disaster	Input post-disaster	$512 \times 512 \times 3$	-
$\text{Encoder}_{1\text{-}1}$	$128 \times 128 \times 64$	$1, 3, 64$	$\text{Encoder}_{21\text{-}1}(f_{1_1})$	$\text{Encoder}_{21\text{-}2}(f_{1_2})$	$128 \times 128 \times 64$	$1, 3, 64$
$\text{Encoder}_{1\text{-}2}$	$64 \times 64 \times 128$	$2, 8, 128$	$\text{Encoder}_{22\text{-}1}(f_{2_1})$	$\text{Encoder}_{22\text{-}2}(f_{2_2})$	$64 \times 64 \times 128$	$2, 8, 128$
$\text{Encoder}_{1\text{-}3}$	$32 \times 32 \times 320$	$5, 27, 320$	$\text{Encoder}_{23\text{-}1}(f_{3_1})$	$\text{Encoder}_{23\text{-}2}(f_{3_2})$	$32 \times 32 \times 320$	$5, 27, 320$
$\text{Encoder}_{1\text{-}4}$	$16 \times 16 \times 512$	$8, 3, 512$	$\text{Encoder}_{24\text{-}1}(f_{4_1})$	$\text{Encoder}_{24\text{-}2}(f_{4_2})$	$16 \times 16 \times 512$	$8, 3, 512$
$\text{Decoder}_{1\text{-}1}$	$16 \times 16 \times 256$	-	$\text{Decoder}_{21\text{-}1}$	$\text{Decoder}_{21\text{-}2}$	$16 \times 16 \times 256$	-
$\text{Decoder}_{1\text{-}2}$	$32 \times 32 \times 128$	-	$\text{Decoder}_{22\text{-}1}$	$\text{Decoder}_{22\text{-}2}$	$32 \times 32 \times 128$	-
$\text{Decoder}_{1\text{-}3}$	$64 \times 64 \times 64$	-	$\text{Decoder}_{23\text{-}1}$	$\text{Decoder}_{23\text{-}2}$	$64 \times 64 \times 64$	-
$\text{Decoder}_{1\text{-}4}$	$128 \times 128 \times 32$	-	$\text{Decoder}_{24\text{-}1}$	$\text{Decoder}_{24\text{-}2}$	$128 \times 128 \times 32$	-
Output	$512 \times 512 \times 2$	-	Output		$512 \times 512 \times 5$	-

ground truth value and the predicted value. The denominator, which represents the sum of the ground truth value and the predicted value, ensures that the loss function is piecewise differentiable, making it trainable. N is the number of pixels in HRS images.

Finally, we generate the overall loss L_{seg} of the first-stage by using L_{dice} and L_{focal}

$$L_{\text{seg}} = L_{\text{dice}} + \lambda_{\text{focal}} L_{\text{focal}} \tag{4}$$

where λ_{focal} is the scaling factor that controls the weight of L_{focal}.

2) Stage 2 (Damage Assessment): During damage assessment, it is extremely critical to compare the changes between pre- and postdisaster images. To achieve this, we propose a dual-branch network to process both pre- and postdisaster images in the second stage, as shown in Fig. 3.

This dual-branch structure employs a weighted shared network to process pre- and postdisaster images separately. The specific parameters of the two-stage method and the dimensions of the feature maps are presented in Table I, where the values in the "Block" columns indicate the number of heads, layers, and channels in different transformer blocks. As illustrated in Fig. 3, the output of the second stage corresponds to the building damage assessment results. To refine the results of building damage assessment obtained in the second stage, segmentation results of building locations obtained in the first stage are utilized, due to the blurred outline of the building in the postdisaster image. The precise formula for this refinement process is as follows:

$$O = O_b \cdot \arg\max_c (O_d(c, h, w)) \tag{5}$$

where $O_d \in \mathbb{R}^{5 \times H \times W}$ represents the output result of damage assessment network in the second stage, $O_b \in \mathbb{R}^{H \times W}$ represents the output result of building segmentation network in the first stage, $\cdot$ represents the pixel-by-pixel multiplication and O represents the final predicted result.

In the training process of the second stage, the network structure related to the first stage is employed in the encoder stage, and the final training parameters of the first stage model are used as the initial parameters for the second-stage model.

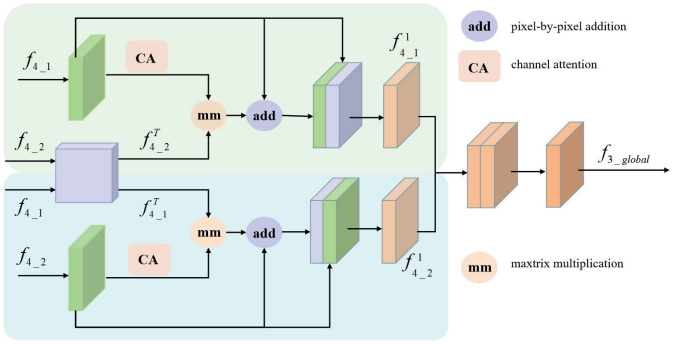


Fig. 4. GF Module. Taking Encoder_{24} in the second stage as an example. The GF module facilitates the fusion of deep features within the dual-branch network. f_{4_1} represents the feature map of the predisaster image obtained from $\text{Encoder}_{24\text{-}1}$, while f_{4_2} represents the feature map of the postdisaster image obtained from $\text{Encoder}_{24\text{-}1}$. f_{3_global} represents the fusion feature obtained from the GF module.

The total loss L_{cls} of second stage of training are composed of cross-entropy loss L_{ce} and building segmentation loss L_{seg} from the first stage. The specific formulas are as follows:

$$L_{\text{ce}}(y, \hat{y}) = -\hat{y}^T \log(\text{softmax}(y)) \tag{6}$$

$$L_{\text{cls}} = L_{\text{seg}} + \lambda_{\text{ce}} L_{\text{ce}} \tag{7}$$

where $y \in \mathbb{R}^{5 \times H \times W}$ represents the output result of damage assessment, $\hat{y} \in \mathbb{R}^{5 \times H \times W}$ represents ground truth, λ_{ce} is the scaling factor that controls the weight of L_{ce}.

B. Guided Fusion

To explore the effective information between pre- and postdisaster images and fully exploit the clear building outlines in predisaster images, we introduce GF module. Previous studies have shown that the deep layers of neural network contain higher semantic features and global expressions, as demonstrated by FPN [37], which integrates high-level and low-level features. Therefore, we propose a dual-branch siamese-transformer network to transfer high-level feature information to low-level feature information.

If feature interaction at different levels is performed on only one branch, the significance of the dual-branch structure used

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

in the second stage of damage assessment may be compromised. To address this issue, we propose a feature transmission medium known as the GF module within the dual-branch structure. This module acts as a guide and facilitates the fusion of the high-level semantic features in the deep layers of the dual-branch network. The GF module implements the process of transferring high-level features to low-level features in a dual-branch network, as depicted in Fig. 4.

To illustrate the GF module, we take Encoder_{24} in the second stage as an example (which includes $\text{Encoder}_{24\text{-}1}$ and $\text{Encoder}_{24\text{-}2}$, as shown in Table I). Specifically, the GF module first calculates the importance of the pre- and postdisaster images to each other, and then fuses the high-level features of the pre- and postdisaster images based on their respective importance weights. The specific formulae are as follows:

$$f_{4_1}^{1} = \text{CB}\left(\left[\text{CA}\left(f_{4_1}\right) \otimes f_{4_2}^{T} + f_{4_1}, f_{4_1}\right]\right) \tag{8}$$

$$f_{4_2}^{1} = \text{CB}\left(\left[\text{CA}\left(f_{4_2}\right) \otimes f_{4_1}^{T} + f_{4_2}, f_{4_2}\right]\right) \tag{9}$$

$$f_{3_\text{global}} = \text{CB}\left(\left[f_{4_1}^{1}, f_{4_2}^{1}\right]\right) \tag{10}$$

where f_{4_1} represents the feature map of the predisaster image from the $\text{Encoder}_{24\text{-}1}$, f_{4_2} represents the feature map of the postdisaster image from the $\text{Encoder}_{24\text{-}2}$. $f_{4_1}^{T}$ and $f_{4_2}^{T}$ represent the transpose of their corresponding features, respectively. $f_{4_1}^{1}$ and $f_{4_2}^{1}$ represent the temporary state of their corresponding features, respectively. $\text{CB}(*)$ represents 3*3 convolution blocks, which reduce the number of channels to half as the input. $\text{CA}(*)$ represents Channel Attention Mechanism. $\otimes$ represents the matrix multiplication operation. $[*, *]$ represents the channel concatenation operation. $+$ means the addition operation.

C. Cross-Scale-Guided Fusion

Once the GF module has completed the fusion process, the features from the pre- and postdisaster images are combined, resulting in a transmitted medium that connects the deep and shallow layers of the dual-branch network. By utilizing this medium, we can achieve the cross-scale feature transfer from bottom to top between dual-branch networks. Therefore, we propose a CSGF module, which fuses the deep features obtained from the GF module with the shallow features of the dual-branch network.

To illustrate the functionality of the CSGF module, we use $\text{Encoder}_{2}3$ in the second stage as an example (which includes Encoder_{23_1} and Encoder_{23_2}). As shown in Fig. 5, we first fuse the feature f_{3_global} obtained from GF module with the features of the current layer f_{3_1} and f_{3_2} in the dual-branch network, resulting in f_{3_csf}. We then use f_{3_csf} as the weight to perform weighted fusion with f_{3_1} and f_{3_2}, respectively, ultimately obtaining f_{2_1} and f_{2_2}. The specific formulae are as follows:

$$f_{3_\text{csf}} = \left(f_{3_1} + f_{3_2}\right) \otimes f_{3_\text{global}}^{T} \tag{11}$$

$$f_{2_1} = \text{CB}\left(f_{3_\text{csf}} \otimes f_{3_1}^{T} + f_{3_1}\right) \tag{12}$$

$$f_{2_2} = \text{CB}\left(f_{3_\text{csf}} \otimes f_{3_2}^{T} + f_{3_2}\right) \tag{13}$$

where f_{3_1} represents the feature map of the predisaster image from the Encoder_{23_1}, f_{3_2} represents the feature map of the postdisaster image from the Encoder_{23_2}. $f_{3_1}^{T}$ and $f_{3_2}^{T}$ represent the transpose of their corresponding features, respectively. f_{3_global} is the output obtained through the GM module and represents the feature transmitted from Encoder_{24} to Encoder_{23}. $f_{3_\text{global}}^{T}$ represents the transpose of the corresponding feature. Other operation symbols are defined in the same manner as the operators in the GF module.

With the CSGF module, we achieve CSGF between features at different scales in a dual-branch network.

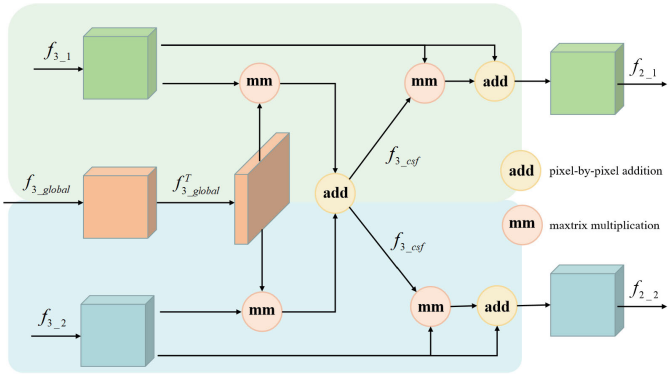


Fig. 5. CSGF Module. Taking Encoder_{23} in the second stage as an example, the CSGF module facilitates the fusion of features between different scales within the dual-branch network. f_{3_1} represents the feature map of the predisaster image from $\text{Encoder}_{23\text{-}1}$, f_{3_2} represents the feature map of the postdisaster image from Encoder_{23_2}, and f_{3_global} represents the output feature map of the GF module. The CSGF module enables cross-layer conduction of features between different layers.

TABLE II
DATA SPLIT AND BUILDING ANNOTATIONS OF XBD DATASET

Split	Number of images	Number of polygons
Train	18336	632228
Test	1866	109724

IV. EXPERIMENT

A. Dataset and Evaluation Metric

1) Dataset: To validate the proposed method, we utilize the xBD [19] dataset, which is currently the largest publicly available HRS images dataset for damage assessment. The dataset includes annotations of the location of buildings in predisaster images and the level of damage to buildings in postdisaster images. It contains a large number of matched pre- and postdisaster images with an image size of 1024 × 1024 and a resolution of 0.8 pixels/m, depicting damage caused by various natural disasters such as hurricanes, tsunamis, wildfires, earthquakes, and more, in numerous regions of the world. The xBD involves more than 800 000 labeled building information, and the extent of damage to buildings is divided into four main categories: no damage, minor damage, major damage, and destroyed. The division of training set and test set in xBD is presented in Table II, in which the number of images of train set and test set accounts for nearly 10:1, and that of polygons is close to 6:1.

Furthermore, as depicted in Fig. 6, there is significant variation in the distribution of buildings with different levels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

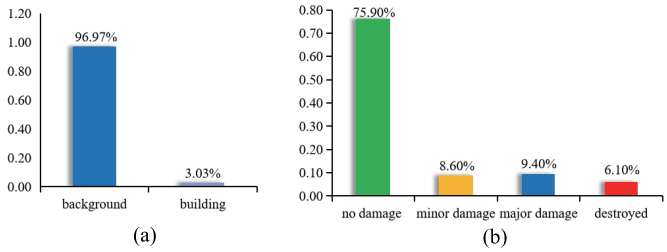


Fig. 6. Statistics of pixels in the xBD dataset. (a) Foreground and background pixel ratio. (b) Pixel ratio of different damage levels.

of damage, with approximately 75.9% of the pixels in the no damaged area. The no damaged level occupies a significantly larger number of pixels than the other damage levels, resulting in a clear long-tailed data distribution. At the same time, there is a big difference in the number of pixels between the background area and the building area. The number of pixels in the background area accounts for nearly 97%.

2) Evaluation Metrics: To evaluate the performance of the proposed method, some quantitative analytical experiments are performed. The building damage assessment consists of two main aspects: the accuracy of building segmentation and building damage assessment. Therefore, $F1_b$ is used to evaluate the accuracy of building segmentation, and harmonic mean $F1_d$ is used to evaluate the accuracy of the damage degree of building assessment. The definitions are as follows:

$$F1_b = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{14}$$

$$F1_d = \frac{n}{\sum_{i=1}^{n} 1/F1_{c_i}} \tag{15}$$

where TP, FP, and FN denote the number of pixels in the segmentation results that are true-positive, false-positive, and false-negative, respectively. $F1_{c_i}$ represents the $F1$ score of the segmentation results of ith level damage degree.

Ultimately, we adopt the challenge metric $F1_s$ used in the xView2 Challenge[2], which is a weighted average of the building segmentation $F1_b$ and the harmonic mean of class-wise damage classification $F1_d$. The formula for this metric is as follows:

$$F1_s = 0.3 \times F1_b + 0.7 \times F1_d. \tag{16}$$

This metric presents a particular challenge, as it penalizes over-fitting to over-presented classes. This is a reasonable metric for evaluating classes that are heavily skewed in the xBD dataset.

3) Experimental Details: All experiments are conducted using the PyTorch framework on 2 Tesla V100 GPUs. During the training process, we randomly crop 512 × 512 sections from the 1024 × 1024 images in the xBD dataset. To alleviate the imbalanced distribution of damage levels in disaster assessment, we employ data augmentation techniques, such as random rotation, erasure, and the addition of Gaussian noise to the randomly cropped images. The Segformer is utilized as the foundational architecture, and the network is initialized with a pretrained model on the ImageNet dataset. For building segmentation, the Adam optimizer is employed with an initial learning rate of 0.0004 for parameter optimization, and epoch-based scheduling is used to adaptively adjust the learning rate during training. The model is trained using a batch size of 32 over 150 epochs for the building segmentation stage. In the damage assessment stage, we use the Adam optimizer to optimize the model with a learning rate of 0.0002, also dynamically adjusted. The damage assessment model is trained with a batch size of 32 for a total of 30 epochs.

[2]https://xview2.org

B. Result Analysis

We compare our method with the one- and two-stage methods on the xBD dataset, and the results are presented in Table III (The results of comparison methods are from the original papers for undisclosed code methods, while the results for the disclosed code are from the replicated results). To evaluate the effectiveness of our method, we conduct a detailed analysis from both quantitative and visual perspectives.

We begin by comparing our proposed approach to the baseline method of the xBD dataset, which is a two-stage method described in Gupta et al. [19]. In the first stage, the xBD baseline method extracts building locations using the SpaceNet [49]. Then, in the second stage, the method classifies the buildings using ResNet and a shallow CNN. However, the xBD baseline method performs poorly in assessing minor and major damage in the second stage, resulting in a low overall score of $F1_s$.

Next, we compare recent major one-stage methods based on change detection. W-Net [44] and U-Net++ [45] are two typical change detection methods for disaster assessment. They compare the differences between predisaster and postdisaster images using siamese networks with weight sharing. Weber and Kané [15] enhance the integration of features by introducing FPN structures into the network based on the change detection framework. ChangeOS [46] ensures the consistency of damage assessment of the building areas through postprocessing based on the change detection framework. RescueNet [47] improves the information interaction between the predisaster images and the postdisaster images by integrating deep features of the siamese network at multiple scales in a one-stage method. BSSnet [17] improves building segmentation by introducing contrastive loss and boundary loss into the one-stage method. Although the one-stage method can achieve the end-to-end process, it is generally less effective than the two-stage method in building segmentation.

Finally, we compare the performance of our method with other two-stage methods. The 1st and 2nd methods in xView2 employ a two-stage approach similar to the baseline method; however, during the second stage, they utilize a semantic segmentation network to evaluate the damage of buildings, thereby significantly exceeding the performance of the baseline method. Based on the 1st method in xView2, Deng and Wang [16] introduce an attention mechanism in the second stage to enhance the information fusion between predisaster and disaster images, which improves the performance of building damage assessment.

1) Quantitative Analysis: As shown in Table III, one-stage methods like W-Net [44] and U-Net++ [45] have lower

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

TABLE III
RESULTS OF DAMAGE ASSESSMENT ON XBD DATASET

Type	Method	$F1_s$	$F1_b$	$F1_d$	No damaged	Minor damaged	Major damaged	Destroyed
One stage	Hao et al. [18]	0.719	0.730	0.714	0.923	0.398	0.675	0.812
	WNet [44]	0.737	0.817	0.703	0.884	0.518	0.684	0.855
	U-Net++ [45]	0.740	0.819	0.707	0.886	0.523	0.689	0.858
	Weber et al. [15]	0.741	0.835	0.700	0.906	0.493	0.722	0.837
	ChangeOS [46]	0.755	0.857	0.711	0.891	0.531	0.724	0.808
	RescueNet [47]	0.770	0.840	0.735	0.885	0.563	0.771	0.808
	DamFormer [48]	0.770	0.869	0.728	0.899	0.568	0.726	0.805
	BSSNet [17]	0.772	0.863	0.733	0.904	0.533	**0.772**	0.846
Two stage	xBD Baseline [19]	0.384	0.852	0.184	0.663	0.144	0.090	0.466
	xView2 1st∗	0.787	0.861	0.755	0.919	0.572	0.770	0.863
	xView2 2nd⋆	0.768	0.840	0.737	0.928	0.538	0.752	0.859
	Deng et al. [16]	0.789	0.874	0.753	0.952	0.577	0.753	0.833
	Ours	**0.806**	**0.885**	**0.772**	**0.954**	**0.600**	0.760	**0.869**

∗ https://github.com/DIUx-xView/xView2_first_place
⋆ https://github.com/DIUx-xView/xView2_second_place

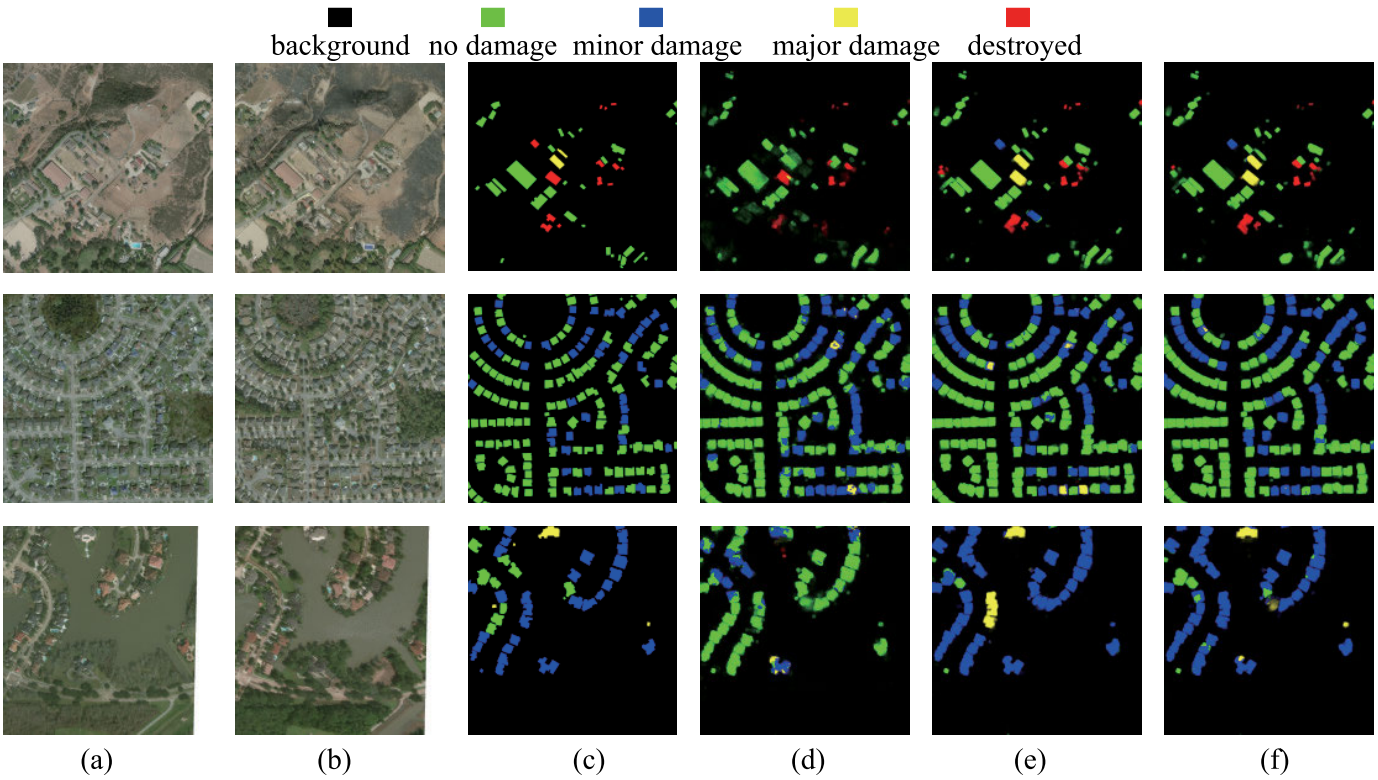


Fig. 7. Results of damage assessment. (Left to right) (a) Predisaster image. (b) Postdisaster image. (c) Ground truth., (d) U-Net++. (e) xView2 1st. (f) Our method. Our method achieves significantly better performances in assessing building damage of no damage and minor damage.

performance in building segmentation and disaster assessment compared to the two-stage methods. However, they can achieve end-to-end training and inference. Table III indicates that our method achieves the best building segmentation results among all the compared methods. Specifically, our method achieves 0.885 in the segmentation evaluation $F1_b$, which is 1.3% higher than the second-best method. Our method, which adopts a hierarchical transformer structure, outperforms compared methods in building segmentation, demonstrating the effectiveness of our approach to separately extracting building locations in the first stage. In terms of damage assessment, our method achieves 0.772 in $F1_d$, which is higher than the second-best method of 2.3%, and our method achieves the best performance in the three damage levels of no damaged, minor damaged, and destroyed. The improvement of our method in damage assessment is attributed to the proposed GF and CSGF modules, which we demonstrate in the ablation experiment.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RESULTS OF DAMAGE ASSESSMENT UNDER DIFFERENT TYPES OF DISASTERS

Type	$F1_b$		No damage		Minor damage		Major damage		Destroyed	
	xView2 1st	our	xView2 1st	our	xView2 1st	our	xView2 1st	our	xView2 1st	our
Guatemala Volcano	0.4278	**0.6289**	0.9069	**0.9164**	0.1735	**0.3190**	0	0	0.8601	**0.8602**
Hurricane Florence	0.9097	**0.9131**	**0.9860**	0.9844	0.3854	**0.4169**	0.8733	**0.8763**	**0.0777**	0.0565
Hurricane Harvery	0.8869	**0.8906**	**0.9195**	0.8826	**0.4832**	0.4562	0.8745	**0.8772**	0.1495	**0.1886**
Hurricane Mattew	0.7845	**0.7942**	0.4065	**0.4665**	0.8082	**0.8542**	**0.3690**	0.2869	0.6654	**0.6918**
Hurricane Michael	0.8587	**0.8656**	0.8250	**0.8317**	0.5351	**0.5593**	**0.4663**	0.4541	0.5145	**0.5837**
Mexio Earthquake	0.8446	**0.8492**	0.9937	**0.9937**	0	0	0	0	0	0
Midwest Flooding	0.8832	**0.8891**	0.9822	**0.9829**	**0.0725**	0.0361	**0.4506**	0.4228	**0.1964**	0.0742
Palu Tsunami	0.8995	**0.9027**	0.9808	**0.9840**	0	0	**0.3764**	0.3245	**0.9192**	0.9022
Santa Rosa wildfire	0.8841	**0.8959**	0.9932	**0.9940**	0.0175	**0.1137**	**0.1299**	0.0055	0.9871	**0.9878**
Socal Fire	0.8508	**0.8587**	0.9874	**0.9877**	0.0002	**0.0073**	**0.1842**	0.0976	0.8775	**0.8882**

TABLE V
ABLATION STUDY OF TWO PROPOSED MODULES

Method	$F1_s$	$F1_b$	$F1_d$	No damage	Minor damage	Major damage	Destroyed
vanilla	0.769	0.885	0.719	**0.955**	0.526	**0.765**	0.765
vanilla + GF	0.781	0.885	0.738	0.943	0.521	0.705	0.812
vanilla + CSGF	0.788	0.885	0.747	0.948	0.561	0.730	**0.871**
vanilla + GF + CSGF	**0.806**	**0.885**	**0.772**	0.954	**0.600**	0.760	0.869

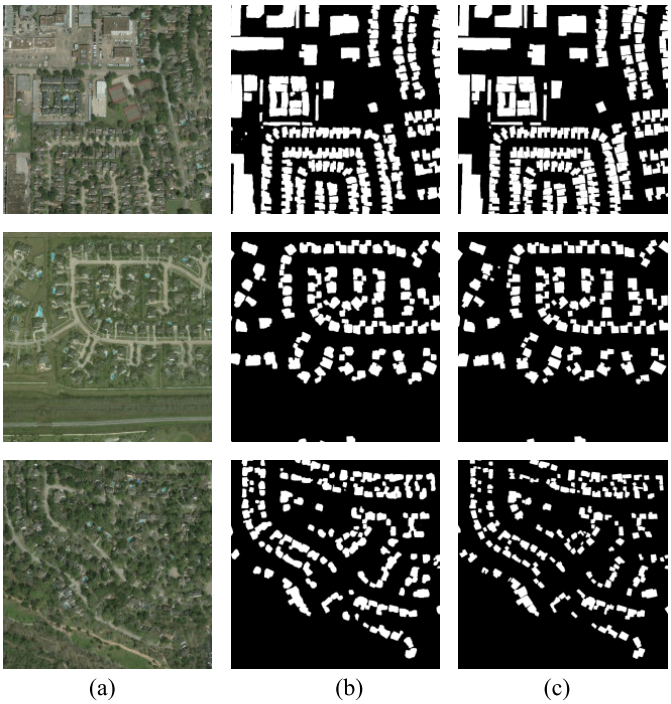


Fig. 8. Examples of building segmentation in the first stage. (a) Predisaster. (b) Prediction. (c) Ground truth.

But our method also has shortcomings, such as $F1$ being lower than RescueNet, BSSNet, and xView2 1st method on major damages. Since the xBD dataset itself suffers from severe class imbalance, it is difficult to fully address the problem of data distribution from the model. Notably, our method significantly improves the performance of the minor damage assessment, with a 4.0% improvement compared to the second-best method.

TABLE VI
ABLATION STUDY OF PARAMETERS AND INFERENCE TIME

Method	Params(M)	FLOPs(G)	Inference(ms/img)	FPS
vanilla	**47.34**	**84.67**	**41.60**	**23.97**
vanilla+GF	53.17	94.99	43.26	23.12
vanilla+CSGF	49.58	88.91	42.12	23.74
vanilla+GF+CSGF	54.51	98.45	61.87	16.16

Then, we validate the performance of our method for each type of natural disaster individually, as shown in Table IV. Our method outperforms the xView2 1st method in terms of building segmentation for each type of disaster. In terms of damage assessment, our method performs significantly outperforms xView2 1st method in the three levels of no damage, minor damage, and destroyed. However, for major damage, the performance of our method declines somewhat, possibly due to over-fitting caused by the limited number of major damage samples. Future studies may consider increasing the number of major damage samples by data augmentation to improve the generalization of the proposed method.

2) Visual Analysis: We present examples of damage assessment in Fig. 7, where our method demonstrates significant

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

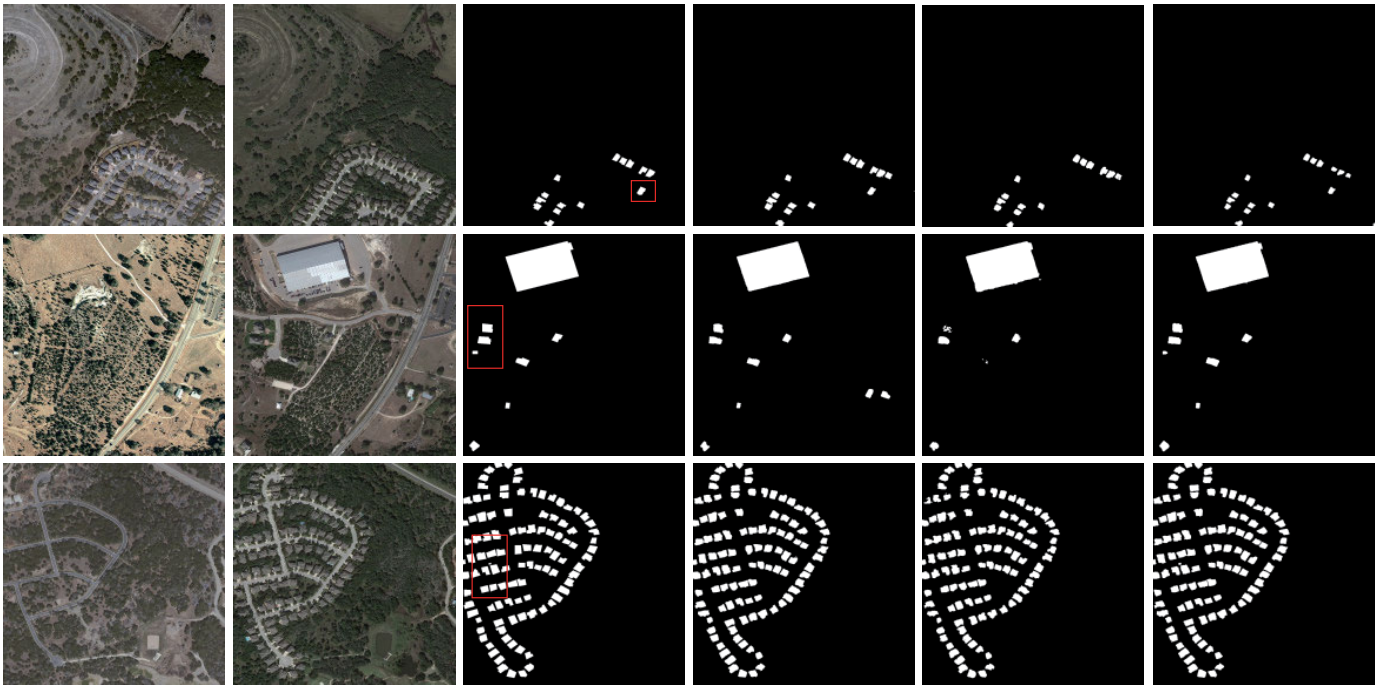


Fig. 9. Results of building change detection on LEVIR-CD dataset. (Left to right) (a) Before change. (b) After change. (c) Ground truth. (d) xView2 1st. (e) U-Net++. (f) Our method.

improvements over the xView2 1st method in identifying no damage, minor damage, and destroyed buildings. Specifically, in Fig. 7, our method successfully distinguishes subtle differences between no damage and minor damage, while the xView2 1st method often misclassifies minor damage as major damage. These findings are further supported by the quantitative analysis of no damage and minor damage presented in Table III, which validates the superiority of our proposed method.

In addition, we also demonstrate the effectiveness of our method in building segmentation from the perspective of visualization. Fig. 8 displays several examples of building segmentation, where our predictions are extremely close to the ground truth.

C. Ablation Study

In the ablation study, we conduct three experiments: module ablation experiments, generalization experiments for each type of disaster, and parameter and inference time ablation experiments.

Table V presents the results of the module ablation experiments on the GF and CSGF modules. With the GF module, which integrates the dual-branch features, we observe a modest 1.6% improvement in damage assessment performance. By combining the simple concatenation and CSGF modules leads to a 2.4% boost in damage assessment performance. When both the GF and CSGF modules are used, our approach achieves the best performance, with a remarkable 4.8% improvement over the vanilla model.

In addition, we conduct an evaluate of the inference time, number of parameters, and computational cost (floating-point operations per second, FLOPs) of our model, and the results are presented in Table VI. Table VI shows a slight increase in the number of parameters for both the GF and CSGF modules compared to the vanilla model. This increase also leads to a small rise in the inference time [frames per second (FPS)], but the overall inference time remains reasonable.

TABLE VII
GENERATION EXPERIMENT ON THE LEVIR-CD DATASET

Method	Precision(%)	Recall(%)	F1(%)	IoU(%)
U-Net++	85.96	66.45	74.96	59.11
xView2 1st(Stage 2)	86.31	72.19	78.62	66.23
Ours(Stage 2)	**89.76**	**87.17**	**88.45**	**79.13**

D. Generalization Experiment

To verify the generalization of our proposed method, we conduct experiments on the LEVIR-CD [50] dataset, which is primarily used for change detection. We select the change detection method of U-Net++ and the building damage assessment method of the xView2 1st as benchmarks for comparison. For the change detection task, our proposed method uses only the second-stage network structure for training. The experimental results, presented in Table VII, demonstrate that our proposed method outperforms the change detection and disaster assessment methods in all four evaluation metrics. Moreover, the visualized results in Fig. 9 show that our proposed method is able to avoid missing detection to a significant extent.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this article, we propose a novel two-stage transformer architecture for building damage assessment. In the first stage, we adopt a hierarchical transformer structure to improve the segmentation performance of building locations. In the second stage, we propose a dual-branch hierarchical transformer that introduces the cross-scale fusion mechanism between pre- and postdisaster images, thereby improving the performance of building damage assessment. Specifically, our method introduces two modules, GF and CSGF, to achieve cross-scale fusion between same-scale and multiscale features in the second stage. To alleviate the class imbalance in damage assessment, we employ multiple data augmentations and loss optimization strategies. The experimental results on the xBD dataset demonstrate that our method achieves SOTA performance in building damage assessment.

References

[1] M. K. Van Aalst, "The impacts of climate change on the risk of natural disasters," *Disasters*, vol. 30, no. 1, pp. 5–18, Mar. 2006.

[2] K. Amadeo, "Hurricane harvey facts, damage and costs," Jul. 2018. [Online]. Available: https://www.thebalancemoney.com/hurricane-harvey-facts-damage-costs-4150087

[3] J.-Y. Rau, L.-C. Chen, J.-K. Liu, and T.-H. Wu, "Dynamics monitoring and disaster assessment for watershed management using time-series satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1641–1649, Jun. 2007.

[4] N. Pierdicca et al., "Triple collocation to assess classification accuracy without a ground truth in case of earthquake damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 485–496, Jan. 2018.

[5] B. Peng, Q. Huang, and J. Rao, "Spatiotemporal contrastive representation learning for building damage classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 8562–8565.

[6] S. Zhai, G. Song, X. Ye, F. Yu, and N. He, "Assessment of drought occurrence in Bei-bu wheat districts, China between 1960 and 2012," in *Proc. 25th Int. Conf. Geoinformatics*, Aug. 2017, pp. 1–5.

[7] D. H. A. Al-Khudhairy, I. Caravaggi, and S. Giada, "Structural damage assessments from Ikonos data using change detection, object-oriented segmentation, and classification techniques," *Photogramm. Eng. Remote Sens.*, vol. 71, no. 7, pp. 825–837, Jul. 2005.

[8] J.-F. Zhang, L.-L. Xie, and X.-X. Tao, "Change detection of earthquake-damaged buildings on remote sensing image and its application in seismic disaster assessment," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2003, pp. 2436–2438.

[9] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[10] W. He, N. Yokoya, L. Yuan, and Q. Zhao, "Remote sensing image reconstruction using tensor ring completion and total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8998–9009, Nov. 2019.

[11] Z. Hong et al., "Classification of building damage using a novel convolutional neural network based on post-disaster aerial images," *Sensors*, vol. 22, no. 15, p. 5920, Aug. 2022.

[12] X. Ye et al., "Building-based damage detection from postquake image using multiple-feature analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 499–503, Apr. 2017.

[13] T. Ci, Z. Liu, and Y. Wang, "Assessment of the degree of building damage caused by disaster using convolutional neural networks in combination with ordinal regression," *Remote Sens.*, vol. 11, no. 23, p. 2858, Dec. 2019.

[14] Y. Zhang, S. Zhang, Y. Li, and Y. Zhang, "Coarse-to-fine satellite images change detection framework via boundary-aware attentive network," *Sensors*, vol. 20, no. 23, p. 6735, Nov. 2020.

[15] E. Weber and H. Kané, "Building disaster damage assessment in satellite imagery with multi-temporal fusion," 2020, *arXiv:2004.05525*.

[16] L. Deng and Y. Wang, "Post-disaster building damage assessment based on improved U-Net," *Sci. Rep.*, vol. 12, no. 1, p. 15862, Sep. 2022.

[17] H. Xie, X. Hu, H. Jiang, and J. Zhang, "BSSNet: Building subclass segmentation from satellite images using boundary guidance and contrastive learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7700–7711, 2022.

[18] H. Hao et al., "An attention-based system for damage assessment using satellite imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 4396–4399.

[19] R. Gupta et al., "Creating xBD: A dataset for assessing building damage from satellite imagery," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 10–17.

[20] Q. Dung Cao and Y. Choe, "Post-hurricane damage assessment using satellite imagery and geolocation features," 2020, *arXiv:2012.08624*.

[21] J. Tian, S. Cui, and P. Reinartz, "Building change detection based on satellite stereo imagery and digital surface models," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 406–417, Jan. 2014.

[22] Y. Bai et al., "Pyramid pooling module-based semi-Siamese network: A benchmark model for assessing building damage from xBD satellite imagery datasets," *Remote Sens.*, vol. 12, no. 24, p. 4055, Dec. 2020.

[23] Y. Shen et al., "BDANet: Multiscale convolutional neural network with cross-directional attention for building damage assessment from satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5402114.

[24] A. Vaswani et al., "Attention is all you need," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, Mar. 2017, pp. 1–12.

[25] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[26] N. Carion et al., "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jul. 2020, pp. 213–229.

[27] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[28] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.

[29] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6877–6886.

[30] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[31] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 548–558.

[32] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2005, pp. 886–893.

[33] D. G. Lowe, "Object recognition from local scale-invariant features," in *Proc. 7th IEEE Int. Conf. Comput. Vis.*, Dec. 1999, pp. 1150–1157.

[34] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 2564–2571.

[35] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[36] F. Iandola, M. Moskewicz, S. Karayev, R. Girshick, T. Darrell, and K. Keutzer, "DenseNet: Implementing efficient ConvNet descriptor pyramids," 2014, *arXiv:1404.1869*.

[37] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[38] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[39] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[40] M. Jaderberg, K. Simonyan, and A. Zisserman, "Spatial transformer networks," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, vol. 28, Nov. 2015, pp. 1–15.

[41] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[42] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.

[43] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.

[44] B. Hou, Q. Liu, H. Wang, and Y. Wang, "From W-Net to CDGAN: Bitemporal change detection via deep learning techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1790–1802, Mar. 2020.

[45] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.

[46] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[47] R. Gupta and M. Shah, "RescueNet: Joint building segmentation and damage assessment from satellite imagery," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 4405–4411.

[48] H. Chen, E. Nemni, S. Vallecorsa, X. Li, C. Wu, and L. Bromley, "Dual-tasks Siamese transformer framework for building damage assessment," 2022, *arXiv:2201.10953*.

[49] A. Van Etten, D. Lindenbaum, and T. M. Bacastow, "SpaceNet: A remote sensing dataset and challenge series," 2018, *arXiv:1807.01232*.

[50] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.



Weiwei Xiao (Graduate Student Member, IEEE) received the B.E. degree from the Shenyang University of Chemical Technology, Shenyang, China, in 2018, and the M.S. degree from Northeastern University, Shenyang, in 2022. He is currently pursuing the Ph.D. degree with Harbin Institute of Technology, Shenzhen, China.

His research interests include computer vision application, machine learning, and few-shot learning.



Jingyong Su received the B.E. and M.S. degrees in electrical engineering from the Harbin Institute of Technology, Harbin, China, in 2006 and 2008, respectively, and the Ph.D. degree in statistics from Florida State University, Tallahassee, FL, USA, in 2013.

He is currently a Professor in computer science with the Harbin Institute of Technology, Shenzhen, China. He joined the Department of Mathematics and Statistics, Texas Tech University, Lubbock, TX, USA, in 2013, as an Assistant Professor and became tenured in 2019. His research interests include computer vision, medical image analysis, and functional and shape data analysis.



Yongyong Chen (Member, IEEE) received the B.S. and M.S. degrees from the College of Mathematics of Systems Science, Shandong University of Science and Technology, Qingdao, China, in 2014 and 2017, respectively, and the Ph.D. degree from the University of Macau (Zhuhai), Macau, China, in 2020.

He is currently an Assistant Professor with the Harbin Institute of Technology, Shenzhen, China. His research interests include image processing, data mining, and computer vision.



Guofeng Cao received the B.S. degree in earth sciences from Zhejiang University, Hangzhou, China, 2001, the M.S. degree in cartography and geographic information system (GIS) from the Chinese Academy of Sciences, Beijing, China, 2004, and the Ph.D. degree in geography from the University of California, Santa Barbara, CA, USA 2011.

He is currently an Assistant Professor with the University of Colorado Boulder, Boulder, CO, USA. His research interests include GIScience, geostatistics, GeoAI, remote sensing, big spatiotemporal data analytics, and the applications in natural hazards and environmental health.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:07:38 UTC from IEEE Xplore. Restrictions apply.